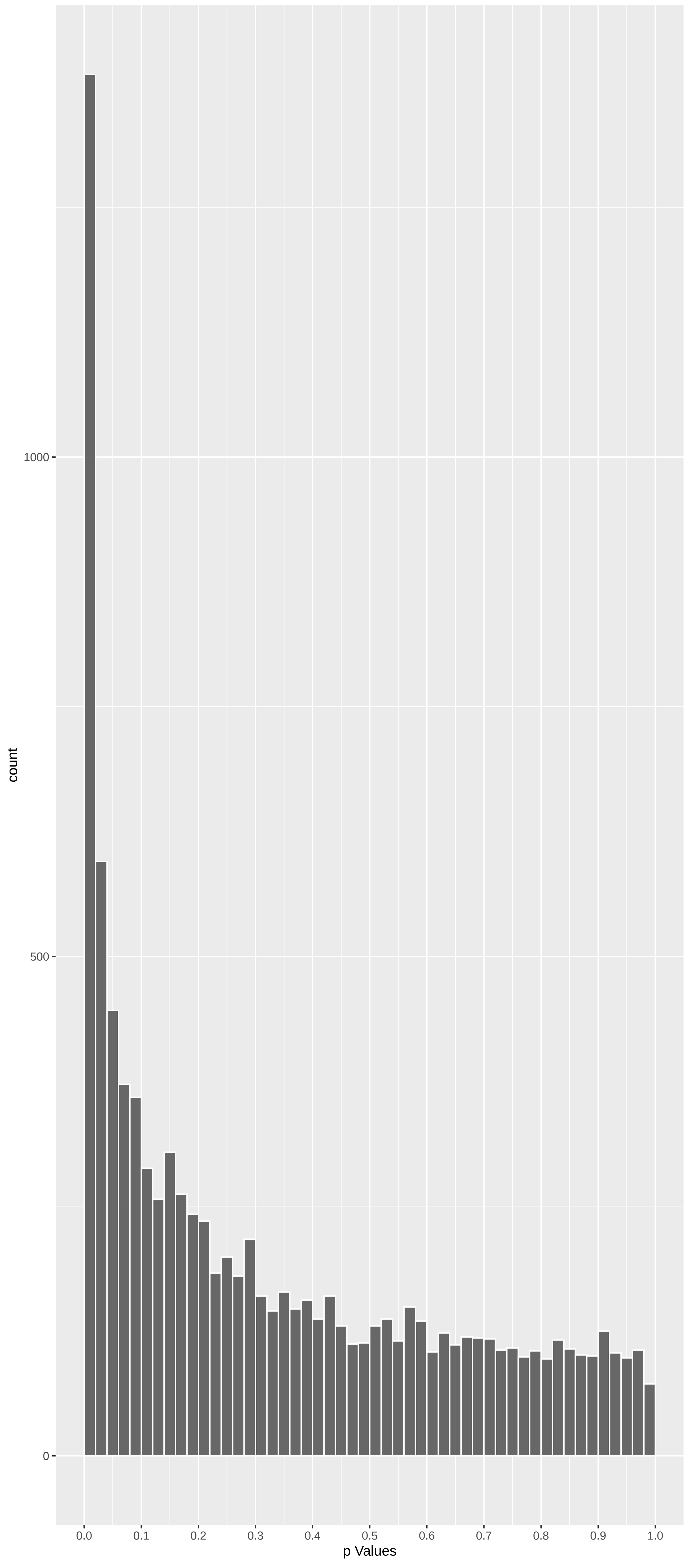
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
PKD1L1	−4.749702	2.037163e−06	4.567e−03	9.998e−01
ANO5	4.914841	8.885438e−07	4.567e−03	9.998e−01
SULT1A2	4.674515	2.946488e−06	4.567e−03	9.998e−01
R3HDM4	4.911197	9.052198e−07	4.567e−03	9.998e−01
H3−3A	4.869563	1.118451e−06	4.567e−03	9.998e−01
DYNLL1	4.727881	2.268749e−06	4.567e−03	9.998e−01
PROM1	4.830790	1.359925e−06	4.567e−03	9.998e−01
TRA2B	4.681926	2.841918e−06	4.567e−03	9.998e−01
TANGO2	4.669210	3.023607e−06	4.567e−03	9.998e−01
RPL23	4.697288	2.636384e−06	4.567e−03	9.998e−01
PPP4C	4.565993	4.971364e−06	6.257e−03	9.998e−01
IQGAP3	−4.569910	4.879337e−06	6.257e−03	9.998e−01
TUBB8	4.541993	5.572502e−06	6.474e−03	9.998e−01
SPINT4	4.512631	6.402824e−06	6.766e−03	9.998e−01
TUBB8B	4.500425	6.781766e−06	6.766e−03	9.998e−01
DNMT3B	4.448534	8.645855e−06	6.766e−03	9.998e−01
SLC22A16	4.431815	9.344327e−06	6.766e−03	9.998e−01
GLRA4	−4.478386	7.520956e−06	6.766e−03	9.998e−01
RPS14	4.449829	8.593851e−06	6.766e−03	9.998e−01
TRPV1	4.442224	8.903365e−06	6.766e−03	9.998e−01
SULT1A3	4.430381	9.406659e−06	6.766e−03	9.998e−01
HIRA	4.345270	1.391044e−05	7.084e−03	9.998e−01
CD80	4.354976	1.330817e−05	7.084e−03	9.998e−01
UBE2B	4.385914	1.154997e−05	7.084e−03	9.998e−01
DACH2	−4.356887	1.319252e−05	7.084e−03	9.998e−01

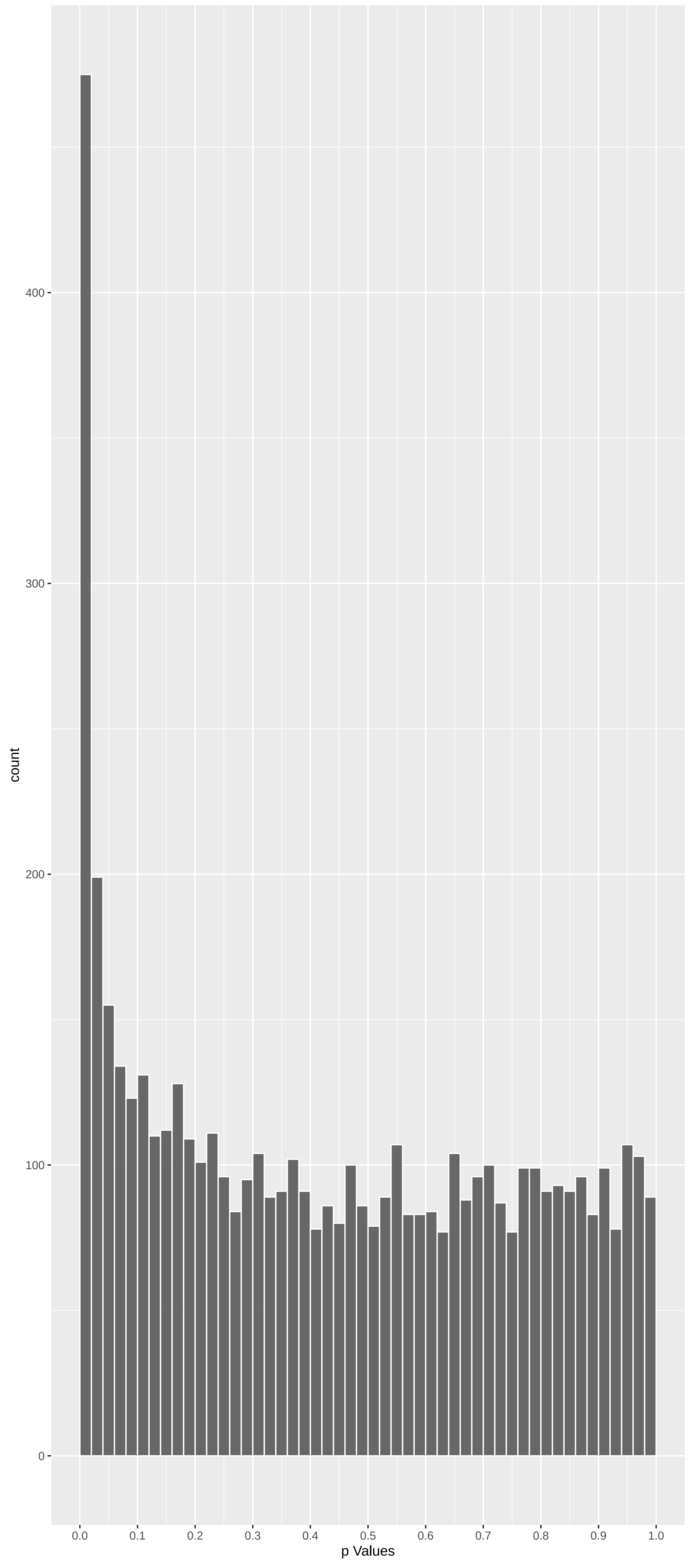
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	0.44354176	6.573739e−01	8.487e−01	9.998e−01
FKBP4	−0.45880854	6.463717e−01	8.435e−01	9.998e−01
CYP26B1	0.55429570	5.793765e−01	8.029e−01	9.998e−01
NDUFAF7	0.28561068	7.751763e−01	9.081e−01	9.998e−01
FUCA2	1.23002567	2.186875e−01	5.084e−01	9.998e−01
DBNDD1	0.57909767	5.625233e−01	7.939e−01	9.998e−01
HS3ST1	1.61816569	1.056269e−01	3.653e−01	9.998e−01
SEMA3F	1.33975623	1.803246e−01	4.664e−01	9.998e−01
CFTR	0.09140354	9.271719e−01	9.717e−01	9.998e−01
CYP51A1	3.90766040	9.319417e−05	1.857e−02	9.998e−01
USP28	−0.78110575	4.347403e−01	7.036e−01	9.998e−01
SLC7A2	1.74908055	8.027710e−02	3.217e−01	9.998e−01
HSPB6	0.84831571	3.962622e−01	6.735e−01	9.998e−01
PDK4	1.70867046	8.751201e−02	3.348e−01	9.998e−01
USH1C	−1.36164403	1.733103e−01	4.614e−01	9.998e−01
BAIAP2L1	1.54804947	1.216104e−01	3.921e−01	9.998e−01
TMEM132A	−2.78746843	5.312163e−03	8.783e−02	9.998e−01
DVL2	−0.43158249	6.660449e−01	8.534e−01	9.998e−01
RPAP3	−1.04831261	2.944946e−01	5.857e−01	9.998e−01
HOXA11	−0.21836838	8.271421e−01	9.313e−01	9.998e−01
TRAPPC6A	0.60202468	5.471577e−01	7.841e−01	9.998e−01
SPATA20	−0.26851032	7.883065e−01	9.143e−01	9.998e−01
CD79B	0.73256680	4.638227e−01	7.265e−01	9.998e−01
RHBDD2	0.01018554	9.918733e−01	9.967e−01	9.998e−01
RPUSD1	1.42592057	1.538913e−01	4.363e−01	9.998e−01

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
ANO5	4.914841	8.885438e−07	4.567e−03	9.998e−01
SULT1A2	4.674515	2.946488e−06	4.567e−03	9.998e−01
R3HDM4	4.911197	9.052198e−07	4.567e−03	9.998e−01
H3−3A	4.869563	1.118451e−06	4.567e−03	9.998e−01
DYNLL1	4.727881	2.268749e−06	4.567e−03	9.998e−01
PROM1	4.830790	1.359925e−06	4.567e−03	9.998e−01
TRA2B	4.681926	2.841918e−06	4.567e−03	9.998e−01
TANGO2	4.669210	3.023607e−06	4.567e−03	9.998e−01
RPL23	4.697288	2.636384e−06	4.567e−03	9.998e−01
PPP4C	4.565993	4.971364e−06	6.257e−03	9.998e−01

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
PKD1L1	−4.749702	2.037163e−06	4.567e−03	9.998e−01
IQGAP3	−4.569910	4.879337e−06	6.257e−03	9.998e−01
GLRA4	−4.478386	7.520956e−06	6.766e−03	9.998e−01
DACH2	−4.356887	1.319252e−05	7.084e−03	9.998e−01
SNRPD1	−4.307108	1.654028e−05	8.059e−03	9.998e−01
MVK	−4.273812	1.921589e−05	8.795e−03	9.998e−01
DUSP27	−4.274691	1.914025e−05	8.795e−03	9.998e−01
SAMD3	−4.234390	2.291727e−05	1.018e−02	9.998e−01
ADGRG4	−4.213349	2.516120e−05	1.056e−02	9.998e−01
GRIN3B	−4.190726	2.780631e−05	1.070e−02	9.998e−01

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Sensory Perception Of Smell (GO:0007608)	−0.23223936	65	9.943e−11	5.356e−07	OR10S1:58 OR6Y1:79 OR51I1:168 OR6K2:196 OR13D1:250 OR2I1P:270
Detection Of Chemical Stimulus Involved	−0.24370719	48	5.323e−09	1.434e−05	OR10S1:58 OR6Y1:79 OR6K2:196 OR13D1:250 OR2I1P:270 OR10D3:300
Detection Of Chemical Stimulus Involved	−0.23956616	47	1.365e−08	2.451e−05	OR10S1:58 OR6Y1:79 OR6K2:196 OR13D1:250 OR2I1P:270 OR10D3:300
Oxidative Phosphorylation (GO:0006119)	0.20274082	54	2.621e−07	3.530e−04	NDUFS5:30 NDUFA11:168 NDUFA12:250 NDUFB5:542 NDUFS3:587 NDUFB3:618
Positive Regulation Of Tyrosine Phosphor	0.21262479	47	4.674e−07	5.036e−04	FLT3:106 TNF:131 IFNG:157 PARP14:217 IL4:224 IL31RA:280
Aerobic Electron Transport Chain (GO:001	0.19618975	52	1.010e−06	6.764e−04	NDUFS5:30 COX5A:322 COX5B:385 NDUFB5:542 NDUFS3:587 NDUFB3:618
Cytokine−Mediated Signaling Pathway (GO:	0.10225147	192	1.130e−06	6.764e−04	SYK:34 CXCL13:86 IL7:94 FLT3:106 TNF:131 CXCR2:174
Mitochondrial ATP Synthesis Coupled Elec	0.19750903	52	8.540e−07	6.764e−04	NDUFS5:30 NDUFA12:250 COX5A:322 COX5B:385 NDUFB5:542 NDUFS3:587
Proton Motive Force−Driven Mitochondrial	0.20652651	47	9.846e−07	6.764e−04	NDUFS5:30 NDUFA11:168 NDUFA12:250 NDUFB5:542 NDUFS3:587 NDUFB3:618
Regulation Of Tyrosine Phosphorylation O	0.18690474	54	2.068e−06	1.114e−03	FLT3:106 TNF:131 IFNG:157 PARP14:217 IL4:224 IL31RA:280
Cilium Assembly (GO:0060271)	−0.09212603	220	2.749e−06	1.346e−03	DYNC2H1:15 CEP164:19 DZIP1L:29 DNAH5:34 KIAA0753:42 OFD1:81
Positive Regulation Of Cytokine Producti	0.08307868	262	4.109e−06	1.844e−03	CD80:23 SYK:34 CD86:39 IL7:94 TNF:131 NLRP9:152
Positive Regulation Of Tumor Necrosis Fa	0.16050771	66	6.663e−06	2.688e−03	SYK:34 CD86:39 CLU:78 IFNG:157 IL33:207 TLR9:310
Proton Motive Force−Driven ATP Synthesis	0.18206815	51	6.986e−06	2.688e−03	NDUFS5:30 NDUFA11:168 NDUFA12:250 NDUFB5:542 NDUFS3:587 NDUFB3:618
Histone Modification (GO:0016570)	0.16616815	60	8.702e−06	3.125e−03	UBE2B:20 GATAD2A:57 RBBP4:77 HDAC11:141 USP3:260 CARM1:319
NADH Dehydrogenase Complex Assembly (GO:	0.17734375	48	2.163e−05	5.578e−03	NDUFS5:30 NDUFA11:168 NDUFB5:542 NDUFS3:587 NDUFB3:618 NDUFA3:724
Aerobic Respiration (GO:0009060)	0.17075755	53	1.736e−05	5.578e−03	NDUFS5:30 NDUFA11:168 NDUFA12:250 NDUFB5:542 NDUFS3:587 NDUFB3:618
Mitochondrial Respiratory Chain Complex	0.17734375	48	2.163e−05	5.578e−03	NDUFS5:30 NDUFA11:168 NDUFB5:542 NDUFS3:587 NDUFB3:618 NDUFA3:724
Mitochondrial Respiratory Chain Complex	0.13957054	77	2.353e−05	5.578e−03	NDUFS5:30 NDUFA11:168 UQCC1:294 NDUFB5:542 NDUFS3:587 NDUFB3:618
Negative Regulation Of Gene Expression (	0.07517488	273	2.117e−05	5.578e−03	ENG:19 ATP2B4:38 TASOR:91 SLFN14:129 TNF:131 IFNG:157
Positive Regulation Of Inflammatory Resp	0.13221176	86	2.321e−05	5.578e−03	TNF:131 IFNG:157 GRN:182 TLR7:194 IL33:207 VAMP8:290
Sphingolipid Biosynthetic Process (GO:00	0.15019864	68	1.887e−05	5.578e−03	ST8SIA5:66 ELOVL7:102 ST8SIA1:136 HACD2:140 SPTLC3:186 PLPP3:195
Steroid Hormone Biosynthetic Process (GO	0.30518912	16	2.381e−05	5.578e−03	HSD17B2:44 CYP17A1:98 HSD17B11:187 CYP19A1:362 HSD17B3:437 HSD3B2:592
Protein Modification Process (GO:0036211	0.05047914	610	2.613e−05	5.864e−03	BCKDK:27 SYK:34 NTAQ1:36 ST8SIA5:66 PPP3CC:71 ST3GAL2:120
Cellular Respiration (GO:0045333)	0.14428674	69	3.482e−05	7.502e−03	NDUFS5:30 NDUFA11:168 NDUFA12:250 COX5A:322 COX5B:385 NDUFB5:542
Glycerophospholipid Metabolic Process (G	0.15772331	56	4.529e−05	9.383e−03	PRDX6:342 GDE1:422 PLA1A:490 SERINC5:609 GPCPD1:785 GDPD1:897
Positive Regulation Of Tumor Necrosis Fa	0.14863499	62	5.270e−05	1.051e−02	SYK:34 CLU:78 IFNG:157 IL33:207 TLR9:310 ORM2:457
Positive Regulation Of Response To Exter	0.09936992	130	9.508e−05	1.829e−02	TNF:131 IFNG:157 GRN:182 TLR7:194 IL33:207 TLR8:233
Cilium Organization (GO:0044782)	−0.07637484	211	1.400e−04	2.600e−02	DYNC2H1:15 CEP164:19 DZIP1L:29 DNAH5:34 KIAA0753:42 OFD1:81
DNA Recombination (GO:0006310)	−0.18033936	36	1.823e−04	2.976e−02	RAG1:73 XRCC5:159 RTEL1:435 RAD50:567 TOP3A:895 MRE11:1168
Cellular Response To Cytokine Stimulus (	0.07341234	222	1.748e−04	2.976e−02	UBE2K:41 IL7:94 FLT3:106 IL31RA:280 IFNAR1:454 RIPK1:522
Histone Ubiquitination (GO:0016574)	0.38250853	8	1.794e−04	2.976e−02	UBE2B:20 UBR2:56 WAC:1350 RNF40:1669 UBE2A:2219 UBE2N:2553
piRNA Processing (GO:0034587)	−0.26377162	17	1.669e−04	2.976e−02	ANKRD34C:21 TDRD12:337 ANKRD34B:528 MOV10L1:615 DDX4:1018 TDRKH:1562
Cell Junction Assembly (GO:0034329)	−0.12027601	80	2.042e−04	3.236e−02	CDC42:70.5 NRCAM:97 RHOA:359 DCHS1:861 PKP3:880 SDK2:937
Regulation Of Dendrite Development (GO:0	−0.17755968	36	2.290e−04	3.473e−02	CHRNB2:290 HECW2:584 KIAA0319:780 SEMA4D:846 COBL:847 RAB17:1098
Steroid Metabolic Process (GO:0008202)	0.13642514	61	2.321e−04	3.473e−02	HSD17B2:44 CYP51A1:52 CYP17A1:98 HSD11B1:156 CYP19A1:362 HSD17B3:437
Positive Regulation Of Interleukin−6 Pro	0.13430675	62	2.586e−04	3.766e−02	SYK:34 TNF:131 IFNG:157 TLR7:194 IL33:207 TLR8:233
Inflammatory Response (GO:0006954)	0.07749122	187	2.713e−04	3.846e−02	CXCL13:86 TPST1:96 TNF:131 IFNG:157 CXCR2:174 GGT5:222
Inorganic Anion Transmembrane Transport	0.13489173	60	3.061e−04	4.122e−02	ANO5:1 SLC12A1:134 SLC20A1:264 SLC1A3:443 ANO6:520 GABRA5:668
Sensory Perception Of Chemical Stimulus	−0.20470531	26	3.043e−04	4.122e−02	CALHM3:68 OR51I1:168 NXNL2:360 OR2S2:384 OR51B5:1278 OR2D2:1451

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	0.07802101	1149	1.407e−18	9.128e−15	R3HDM4:2 SULT1A2:8 DNMT3B:15 UBE2B:20 HIRA:25 HK2:29
BLALOCK_ALZHEIMERS_DISEASE_DN	0.07364929	997	7.349e−15	2.384e−11	DYNLL1:5 UBE2B:20 TMEM50A:37 YPEL5:43 SCFD1:63 ABHD14A:104
REACTOME_OLFACTORY_SIGNALING_PATHWAY	−0.21265644	105	5.421e−14	1.172e−10	OR10S1:58 OR6Y1:79 OR52E8:111 OR51I1:168 OR6K2:196 OR6B2:200
SCHLOSSER_SERUM_RESPONSE_DN	0.09081333	556	3.429e−13	5.561e−10	TRA2B:7 UBR2:56 PPP3CC:71 RBBP4:77 EEF1B2:83 TASOR:91
JOHNSTONE_PARVB_TARGETS_3_DN	0.08015863	690	1.071e−12	1.390e−09	TRA2B:7 NDUFS5:30 TMEM50A:37 ZNF22:111 YES1:123 TFAM:127
DANG_BOUND_BY_MYC	0.07098041	823	7.184e−12	7.768e−09	SYK:34 UBE2K:41 LMX1B:45 RBBP4:77 CIAO1:81 PRMT5:101
WONG_MITOCHONDRIA_GENE_MODULE	0.14111433	192	1.708e−11	1.583e−08	NDUFS5:30 MRPS36:46 TFAM:127 SUCLA2:196 SLC9A5:242 NDUFA12:250
HOUNKPE_HOUSEKEEPING_GENES	0.06693204	860	4.170e−11	3.005e−08	BCKDK:27 TMEM50A:37 UBE2K:41 PHF5A:64 SAV1:69 NAPG:89
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	0.08419630	533	3.830e−11	3.005e−08	ADAM9:47 PI4K2A:51 CYP51A1:52 RBBP4:77 SS18:93 TFAM:127
SPIELMAN_LYMPHOBLAST_EUROPEAN_VS_ASIAN_D	0.08753418	476	7.629e−11	4.949e−08	SYK:34 UBE2K:41 SCFD1:63 TNFSF10:87 PRMT5:101 ZNF22:111
KEGG_OLFACTORY_TRANSDUCTION	−0.17253383	119	8.399e−11	4.953e−08	OR10S1:58 OR6Y1:79 OR52E8:111 OR51I1:168 OR6K2:196 OR6B2:200
MARTORIATI_MDM4_TARGETS_FETAL_LIVER_DN	0.08816743	432	3.997e−10	2.161e−07	FGG:26 TNPO3:33 COL1A1:84 ZNF22:111 DDX21:125 PRKAR2A:165
MARTINEZ_RB1_AND_TP53_TARGETS_UP	0.07934195	529	5.406e−10	2.505e−07	C1orf54:67 RBBP4:77 MYL4:82 SERINC3:114 ST3GAL2:120 BID:122
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.07762150	555	5.165e−10	2.505e−07	CD80:23 SYK:34 CD86:39 TNFSF8:74 CASP3:75 IL7:94
MARSON_BOUND_BY_FOXP3_STIMULATED	0.06215781	854	1.014e−09	4.385e−07	TMEM50A:37 FBXW12:48 MYL4:82 DDB2:88 PRMT5:101 JMJD6:142
BENPORATH_MYC_MAX_TARGETS	0.07261716	611	1.146e−09	4.648e−07	SYK:34 UBE2K:41 SAV1:69 CIAO1:81 PRMT5:101 XK:139
SWEET_LUNG_CANCER_KRAS_UP	0.08980648	390	1.347e−09	5.140e−07	TSPAN8:24 HK2:29 CNIH2:40 PGLS:60 CLU:78 HSPA5:137
PUJANA_BRCA1_PCC_NETWORK	0.05075676	1283	1.772e−09	6.386e−07	RPL23:6 TRA2B:7 HIRA:25 HK2:29 TNPO3:33 SYK:34
REACTOME_METABOLISM_OF_LIPIDS	0.06915451	650	2.355e−09	7.934e−07	HSD17B2:44 PI4K2A:51 CYP51A1:52 CYP17A1:98 ELOVL7:102 HACD2:140
WP_NONALCOHOLIC_FATTY_LIVER_DISEASE	0.15592027	123	2.446e−09	7.934e−07	NDUFS5:30 CASP3:75 BID:122 TNF:131 NDUFA11:168 NDUFA12:250
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	−0.15621381	120	3.558e−09	1.099e−06	GATA5:65 NPBWR2:66 COL20A1:100 FAM217B:124 SLCO4A1:139 LAMA5:231
MOOTHA_MITOCHONDRIA	0.08586631	403	3.888e−09	1.147e−06	BCKDK:27 NDUFS5:30 TNFSF10:87 GLS2:110 BID:122 TFAM:127
WP_NETWORK_MAP_OF_SARSCOV2_SIGNALING_PAT	0.13212013	165	5.042e−09	1.422e−06	FGG:26 CASP3:75 APOD:76 CXCL13:86 TNFSF10:87 IL7:94
REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R	0.14006652	146	5.429e−09	1.428e−06	NDUFS5:30 NDUFA11:168 SUCLA2:196 NDUFA12:250 GLO1:312 IDH3B:318
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.08129798	442	5.504e−09	1.428e−06	ABHD14A:104 SERINC3:114 CALM2:153 NARS1:159 AKR1B1:188 SUCLA2:196
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	0.07082016	580	7.009e−09	1.749e−06	DYNLL1:5 TUBB8:11 TUBB8B:13 UBE2B:20 CD80:23 FGG:26
SHEDDEN_LUNG_CANCER_POOR_SURVIVAL_A6	0.08682666	374	9.341e−09	2.244e−06	HK2:29 AFP:35 NTAQ1:36 UBE2K:41 ORC1:58 TPX2:68
WONG_EMBRYONIC_STEM_CELL_CORE	0.10162905	267	1.198e−08	2.774e−06	PROM1:4 PPP4C:10 MRPS36:46 ORC1:58 PHF5A:64 SS18:93
FLECHNER_BIOPSY_KIDNEY_TRANSPLANT_REJECT	0.07837363	453	1.280e−08	2.862e−06	CYP17A1:98 GRB14:100 YES1:123 NARS1:159 AQP1:189 SLC25A4:214
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.11511903	205	1.431e−08	3.093e−06	TNFSF8:74 CXCL13:86 TNFSF10:87 IL7:94 FLT3:106 LTA:124
REACTOME_SENSORY_PERCEPTION	−0.09679742	291	1.486e−08	3.110e−06	PCDH15:43 SNAP25:47 CALHM1:48 OR10S1:58 CALHM3:68 OR6Y1:79
DODD_NASOPHARYNGEAL_CARCINOMA_UP	−0.04580309	1402	1.593e−08	3.229e−06	GRIN3B:11 DYNC2H1:15 CEP164:19 TTLL5:27 DZIP1L:29 DNAH5:34
DODD_NASOPHARYNGEAL_CARCINOMA_DN	0.05022338	1111	2.461e−08	4.837e−06	TRA2B:7 CD80:23 RPS15A:55 PHF5A:64 TPX2:68 RBBP4:77
MARTINEZ_TP53_TARGETS_UP	0.07148435	524	2.597e−08	4.955e−06	C1orf54:67 RBBP4:77 MYL4:82 SERINC3:114 ST3GAL2:120 BID:122
MARTENS_TRETINOIN_RESPONSE_UP	−0.06471439	643	2.713e−08	5.028e−06	GRIN3B:11 SMTNL2:24 CARD14:84 IRX4:89 SCN5A:92 COL20A1:100
KEGG_OXIDATIVE_PHOSPHORYLATION	0.16731117	92	2.997e−08	5.401e−06	NDUFS5:30 NDUFA11:168 COX5A:322 COX5B:385 COX7B2:483 NDUFB5:542
MOOTHA_PGC	0.08673833	346	3.334e−08	5.846e−06	UBE2B:20 NDUFS5:30 PGLS:60 CIAO1:81 BID:122 HSPA5:137
IBRAHIM_NRF1_UP	0.08447428	357	4.722e−08	7.853e−06	DNMT3B:15 SCPEP1:32 TNPO3:33 GATAD2A:57 SCFD1:63 PRMT5:101
FLECHNER_BIOPSY_KIDNEY_TRANSPLANT_OK_VS_	0.07699776	432	4.713e−08	7.853e−06	DYNLL1:5 SULT1A2:8 TSPAN8:24 SCFD1:63 CLU:78 TNFSF10:87
WINTER_HYPOXIA_METAGENE	0.10840133	210	6.567e−08	1.039e−05	ENG:19 HK2:29 HSPA5:137 RNASEL:158 BIK:258 SLC20A1:264

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Liver carcinoma	0.04957422	2597	9.132e−15	8.954e−11	PROM1:4 DYNLL1:5 DNMT3B:15 SLC22A16:17 SULT1A3:18 ENG:19
Liver neoplasms	0.07000555	1019	1.298e−13	6.366e−10	PROM1:4 DYNLL1:5 DNMT3B:15 UBE2B:20 HK2:29 AFP:35
Neoplasm Metastasis	0.04477332	2913	3.126e−13	1.022e−09	PROM1:4 DYNLL1:5 SULT1A2:8 PPP4C:10 DNMT3B:15 TRPV1:16
Malignant neoplasm of lung	0.05213813	1826	1.172e−12	2.872e−09	PROM1:4 DNMT3B:15 SLC22A16:17 ENG:19 CD80:23 HK2:29
Malignant tumor of colon	0.05507999	1540	2.701e−12	5.297e−09	PROM1:4 TRA2B:7 DNMT3B:15 CD80:23 HK2:29 SYK:34
Primary malignant neoplasm of lung	0.05141115	1701	9.991e−12	1.633e−08	PROM1:4 DNMT3B:15 SLC22A16:17 ENG:19 CD80:23 HK2:29
Carcinoma of lung	0.04910890	1857	1.562e−11	2.188e−08	PROM1:4 DNMT3B:15 SLC22A16:17 ENG:19 CD80:23 HK2:29
Hepatitis C	0.08436194	554	1.840e−11	2.255e−08	UBE2B:20 HK2:29 AFP:35 GGTLC3:49 CASP3:75 CLU:78
Myeloid Leukemia, Chronic	0.07859885	636	2.349e−11	2.559e−08	PROM1:4 DNMT3B:15 CD80:23 SCPEP1:32 SYK:34 CD86:39
Encephalomyelitis	0.11336578	281	7.742e−11	7.591e−08	TRPV1:16 CD80:23 CD86:39 ICOS:53 CASP3:75 CLU:78
Colon Carcinoma	0.04967379	1610	1.307e−10	1.165e−07	PROM1:4 TRA2B:7 DNMT3B:15 ENG:19 CD80:23 TSPAN8:24
Leukemia, Myelocytic, Acute	0.05449868	1285	1.593e−10	1.202e−07	PROM1:4 RPL23:6 DNMT3B:15 SLC22A16:17 CD80:23 SCPEP1:32
Malignant neoplasm of prostate	0.04183549	2444	1.591e−10	1.202e−07	PROM1:4 DYNLL1:5 DNMT3B:15 TRPV1:16 ENG:19 TSPAN8:24
Lupus Erythematosus, Systemic	0.06688328	800	2.426e−10	1.699e−07	PROM1:4 DNMT3B:15 ENG:19 CD80:23 TNPO3:33 SYK:34
Cervix carcinoma	0.06674052	798	2.776e−10	1.701e−07	PROM1:4 DNMT3B:15 ENG:19 UBE2B:20 CD80:23 HK2:29
Psoriasis	0.07628193	602	2.629e−10	1.701e−07	JUND:61 TNFSF8:74 TNFSF10:87 FLT3:106 CMC1:119 LTA:124
Stomach Neoplasms	0.07182800	660	5.150e−10	2.970e−07	PROM1:4 RPL23:6 DNMT3B:15 CD80:23 FGG:26 AFP:35
Fibrosarcoma	0.11656169	240	5.922e−10	3.149e−07	JUND:61 CLU:78 COL1A1:84 TNFSF10:87 PDXP:121 TNF:131
Prostate carcinoma	0.04111841	2343	6.101e−10	3.149e−07	PROM1:4 DYNLL1:5 DNMT3B:15 TRPV1:16 ENG:19 TSPAN8:24
Breast Carcinoma	0.03482745	3758	8.533e−10	4.183e−07	PROM1:4 DYNLL1:5 SULT1A2:8 PPP4C:10 RPS14:14 DNMT3B:15
Hepatitis B	0.07353328	593	1.465e−09	6.840e−07	PROM1:4 CD80:23 AFP:35 RPS15A:55 CASP3:75 TNFSF10:87
Autoimmune Diseases	0.06416270	790	1.554e−09	6.925e−07	RPL23:6 CD80:23 TNPO3:33 AFP:35 CD86:39 GGTLC3:49
Lymphoma	0.05724174	999	1.953e−09	8.326e−07	DYNLL1:5 DNMT3B:15 UBE2B:20 CD80:23 SYK:34 AFP:35
Cholangiocarcinoma	0.09354427	342	3.489e−09	1.425e−06	PROM1:4 DYNLL1:5 DNMT3B:15 HK2:29 AFP:35 CASP3:75
Malignant neoplasm of ovary	0.04651541	1526	4.049e−09	1.588e−06	PROM1:4 DNMT3B:15 ENG:19 HK2:29 SYK:34 HSD17B2:44
Bladder Neoplasm	0.05846943	905	4.639e−09	1.750e−06	TRPV1:16 CD80:23 CD86:39 ADAM9:47 JUND:61 CASP3:75
Carcinoma of bladder	0.05906729	859	7.514e−09	2.540e−06	TRPV1:16 CD80:23 CD86:39 ADAM9:47 JUND:61 TPX2:68
Degenerative polyarthritis	0.06155386	788	7.194e−09	2.540e−06	TRPV1:16 ENG:19 CD86:39 LMX1B:45 JUND:61 COL2A1:72
Systemic Scleroderma	0.08417866	407	7.363e−09	2.540e−06	ENG:19 TNPO3:33 JUND:61 CASP3:75 COL1A1:84 TNFSF10:87
Inflammation	0.09088392	345	8.256e−09	2.698e−06	TRPV1:16 COL2A1:72 APOD:76 CLU:78 TNFSF10:87 TNF:131
Nonalcoholic Steatohepatitis	0.12640993	168	1.742e−08	5.510e−06	DNMT3B:15 CASP3:75 COL1A1:84 TNFSF10:87 CYP17A1:98 TNF:131
Adenocarcinoma of lung (disorder)	0.05711182	865	2.074e−08	6.356e−06	PROM1:4 TRA2B:7 DNMT3B:15 CD80:23 ATAD3C:28 HK2:29
Vascular Diseases	0.09066149	325	2.341e−08	6.956e−06	ENG:19 FGG:26 JUND:61 CASP3:75 CLU:78 DDB2:88
Pneumonia	0.08245225	393	2.563e−08	7.392e−06	DNMT3B:15 FGG:26 SYK:34 CD86:39 ICOS:53 JUND:61
Cystic kidney	−0.18354259	77	2.680e−08	7.509e−06	OFD1:81 LZTFL1:151 LAMA5:231 IFT140:251 HSD17B4:288 DHCR7:374
Thyroid carcinoma	0.06939780	559	2.854e−08	7.772e−06	PROM1:4 GATAD2A:57 JUND:61 CASP3:75 RBBP4:77 CLU:78
Malignant neoplasm of pancreas	0.04547311	1398	3.072e−08	8.140e−06	PROM1:4 TRPV1:16 ENG:19 CD80:23 TSPAN8:24 HK2:29
Endometriosis	0.06663551	598	3.741e−08	9.652e−06	DNMT3B:15 TRPV1:16 HSD17B2:44 SCN11A:62 COL1A1:84 TNFSF10:87
Kidney Failure, Acute	0.10262940	240	4.972e−08	1.096e−05	AFP:35 CLU:78 CALCR:97 TNF:131 GRN:182 AFM:275
cervical cancer	0.06148120	696	4.931e−08	1.096e−05	DNMT3B:15 ENG:19 UBE2B:20 CD80:23 HK2:29 UBE2K:41

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0008469 abnormal protein level	0.06356846	373	4.280e−05	2.033e−02	FGG:26 CD86:39 PI4K2A:51 ICOS:53 COL1A1:84 FLT3:106
MP0002136 abnormal kidney physiology	0.06708868	267	2.199e−04	2.611e−02	TRPV1:16 LMX1B:45 COL2A1:72 SLC12A1:134 IFNG:157 JUNB:184
MP0002429 abnormal blood cell	0.03671908	1130	1.667e−04	2.611e−02	PPP4C:10 DNMT3B:15 ENG:19 CD80:23 FGG:26 CD86:39
MP0002420 abnormal adaptive immunity	0.04117828	818	1.892e−04	2.611e−02	PPP4C:10 CD80:23 FGG:26 CD86:39 JUND:61 CASP3:75
MP0001819 abnormal immune cell	0.03817549	822	5.251e−04	3.362e−02	PPP4C:10 CD80:23 FGG:26 CD86:39 ICOS:53 JUND:61
MP0002086 abnormal extraembryonic tissu	0.05043891	435	4.963e−04	3.362e−02	TRA2B:7 ENG:19 HIRA:25 FGG:26 HSD17B2:44 GATAD2A:57
MP0005000 abnormal immune tolerance	0.06669862	237	5.169e−04	3.362e−02	CD80:23 CD86:39 ICOS:53 TNFSF10:87 LTA:124 TNF:131
MP0002723 abnormal immune serum	0.04345159	595	5.662e−04	3.362e−02	TRPV1:16 CD80:23 FGG:26 CD86:39 TNFSF10:87 FLT3:106
MP0005636 abnormal mineral homeostasis	0.08582401	129	8.597e−04	3.728e−02	CALCR:97 SLC12A1:134 RHCG:326 HK1:377 SLC26A3:410 FTH1:466
MP0009642 abnormal blood homeostasis	0.03220295	1196	7.573e−04	3.728e−02	TRPV1:16 FGG:26 BCKDK:27 AFP:35 PI4K2A:51 ICOS:53
MP0000026 abnormal inner ear	−0.07301129	180	8.633e−04	3.728e−02	PCDH15:43 SCRIB:207 JAG2:219 CDH23:354 OTOF:406 POU3F4:450
MP0009643 abnormal urine homeostasis	0.06489782	218	1.169e−03	4.628e−02	LMX1B:45 SLC12A1:134 IFNG:157 JUNB:184 AQP1:189 MAN2A1:244
MP0003436 decreased susceptibility to	0.15080059	38	1.338e−03	4.889e−02	LTA:124 TNF:131 CXCR2:174 IL1R1:304 CTSS:604 TNFSF13:913
MP0010094 abnormal chromosome stability	−0.12226737	57	1.477e−03	5.010e−02	LIG1:51 RAG1:73 XRCC5:159 PMS2:172 MCPH1:387 NHEJ1:415
MP0000716 abnormal immune system	0.03155580	921	2.716e−03	8.602e−02	PPP4C:10 DNMT3B:15 CD80:23 FGG:26 CD86:39 ICOS:53
MP0000534 abnormal ureter morphology	−0.12176093	46	4.409e−03	1.309e−01	ADAMTS1:712 ETV4:771 UPK2:961 ITGA3:971 SALL1:1175 BMP7:1853
MP0002452 abnormal antigen presenting	0.03683115	538	5.168e−03	1.364e−01	CD80:23 CD86:39 ICOS:53 CASP3:75 COL1A1:84 CXCL13:86
MP0004742 abnormal vestibular system	−0.13334380	37	5.125e−03	1.364e−01	PCDH15:43 LYST:258 CDH23:354 POU3F4:450 HR:594 IREB2:856
MP0001765 abnormal ion homeostasis	0.08069544	99	5.895e−03	1.474e−01	CALCR:97 SLC12A1:134 CALR:306 RHCG:326 ARNTL:407 SLC26A3:410
MP0005330 cardiomyopathy	0.09728097	66	6.539e−03	1.553e−01	JUND:61 TFAM:127 CALR:306 MYOZ2:672 HBEGF:710 CAV3:1077
MP0003656 abnormal erythrocyte physiolo	0.16509837	21	8.920e−03	1.933e−01	MAN2A1:244 PIGA:1094 AQP9:1160 PRKAA1:1270 KLF1:1753 BNIP3L:2097
MP0002080 prenatal lethality	0.02484072	1206	9.177e−03	1.933e−01	TRA2B:7 PPP4C:10 DNMT3B:15 ENG:19 HIRA:25 HK2:29
MP0001533 abnormal skeleton physiology	0.07579649	99	9.698e−03	1.933e−01	TNFSF10:87 LTA:124 TNF:131 CXCR2:174 JUNB:184 IL1R1:304
MP0005319 abnormal enzyme/ coenzyme	0.05873034	165	1.017e−02	1.933e−01	PI4K2A:51 COL1A1:84 TFAM:127 SLC12A1:134 IFNG:157 IL4:224
MP0005025 abnormal response to	0.04181221	339	1.002e−02	1.933e−01	FGG:26 ICOS:53 TNFSF10:87 FLT3:106 LTA:124 TNF:131
MP0005310 abnormal salivary gland	0.12020742	34	1.554e−02	2.839e−01	ST14:166 IL4:224 CTSS:604 TGFB1:613 E2F1:729 TRPC1:948
MP0005332 abnormal amino acid	0.06541439	113	1.725e−02	2.878e−01	BCKDK:27 SLC12A1:134 SLC25A4:214 ACE:403 CD2AP:595 SLC7A9:605
MP0000001 mammalian phenotype	0.01948391	2790	1.735e−02	2.878e−01	PROM1:4 TRA2B:7 PPP4C:10 DNMT3B:15 TRPV1:16 ENG:19
MP0005390 skeleton phenotype	0.05122230	186	1.757e−02	2.878e−01	DNMT3B:15 LMX1B:45 COL1A1:84 TNFSF10:87 LTA:124 TNF:131
MP0000689 abnormal spleen morphology	0.03161499	510	1.906e−02	3.018e−01	CD80:23 CD86:39 COL1A1:84 CXCL13:86 IL7:94 GRB14:100
MP0001845 abnormal inflammatory respons	0.03131833	475	2.446e−02	3.133e−01	TRPV1:16 CD80:23 JUND:61 CASP3:75 TNFSF10:87 BID:122
MP0000015 abnormal ear pigmentation	−0.16852738	15	2.398e−02	3.133e−01	LYST:258 CBL:1353 POLH:1809 HPS4:2069 SLC24A5:2167 HPS5:2906
MP0001800 abnormal humoral immune	0.03870815	315	2.123e−02	3.133e−01	CD80:23 CD86:39 TNFSF10:87 LTA:124 TNF:131 IFNG:157
MP0000049 abnormal middle ear	−0.08974244	54	2.309e−02	3.133e−01	HMX1:164 POU3F4:450 CHD7:494 FOXC2:505 FGF8:903 DLX2:989
MP0010386 abnormal urinary bladder	−0.17311142	14	2.506e−02	3.133e−01	CHRNB2:290 UPK2:961 CHRNB4:1458 MYLK:1506 HTR3A:2030 AGA:2404
MP0003221 abnormal cardiomyocyte apopto	0.11406987	33	2.368e−02	3.133e−01	TFAM:127 CXADR:154 CALR:306 HDAC1:740 IVNS1ABP:1547 PNPLA2:1908
MP0002088 abnormal embryonic growth/wei	0.03806600	308	2.496e−02	3.133e−01	DNMT3B:15 ENG:19 HIRA:25 HK2:29 HSD17B2:44 SS18:93
MP0005166 decreased susceptibility to	0.07852104	70	2.384e−02	3.133e−01	JUND:61 BID:122 IFNG:157 IL4:224 ACE:403 CAMK2D:477
MP0003724 increased susceptibility to	0.16436736	15	2.769e−02	3.372e−01	TNFSF10:87 IL1RN:386 SOD3:978 IL6ST:1039 CD200:1212 DUSP1:3471
MP0000598 abnormal liver morphology	0.03034710	443	3.462e−02	3.574e−01	DNMT3B:15 BCKDK:27 JUND:61 SAV1:69 COL1A1:84 GRB14:100

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
testis	−0.03623133	1338	3.845e−05	2.076e−03	PKD1L1:1 FSHR:16 BNC1:20 TTLL5:27 SPPL2C:49 LRRC27:62
EBV.lymphocyte	0.04609504	652	1.055e−04	2.849e−03	CD80:23 HK2:29 SYK:34 CD86:39 ORC1:58 TPX2:68
liver	0.04359098	321	8.268e−03	1.141e−01	FGG:26 AFP:35 HSD17B2:44 GRB14:100 HSD11B1:156 SLC38A4:191
pancreas	0.07938145	93	8.449e−03	1.141e−01	FBXW12:48 CELA3A:59 CATSPERB:70 TEX11:516 TMED6:627 CLPS:736
amygdala	−0.23307444	8	2.248e−02	2.287e−01	PCDH15:43 PTPRZ1:253 BCAN:385 SLC38A8:3256 NEUROD6:4338 TNR:4503
blood	0.03972722	265	2.807e−02	2.287e−01	R3HDM4:2 IFNG:157 CXCR2:174 TLR8:233 IL18RAP:238 FKBP8:397
endocervix	−0.07368065	70	3.364e−02	2.287e−01	PWP2:174 ZNF516:271 NXNL2:360 STRA6:454 SLC18A2:531 BDKRB1:563
fallopian.tube	−0.08871057	48	3.389e−02	2.287e−01	ADGRG4:9 GATA5:65 TCF23:275 CCDC17:521 DTHD1:556 STOML3:625
cerebellum	−0.03045096	354	5.330e−02	2.905e−01	GLRA4:3 SSTR4:86 TRANK1:122 PCLO:135 SPHKAP:235 TMEM132A:247
ovary	−0.05542389	98	5.918e−02	2.905e−01	CCDC68:142 TCF23:275 CDH23:354 MMP21:365 GREB1:381 GXYLT2:492
tibial.nerve	−0.04096717	183	5.826e−02	2.905e−01	DMD:75 SPTBN5:157 OR6B2:200 SORCS1:269 DCLK3:389 MYOC:451
sigmoid.colon	−0.07445149	50	6.924e−02	3.116e−01	MAB21L2:45 ACTG2:70.5 TACR2:129 HMCN2:468 PLA2G2C:517 PLEKHO1:893
esophagus.mucosa	−0.03405370	213	9.018e−02	3.746e−01	IQANK1:17 BNC1:20 FAM83H:44 KRT78:52 EVPL:91 MROH6:204
hippocampus.proper	0.15134268	9	1.160e−01	4.476e−01	CNIH2:40 FEZF2:2328 NEUROD2:3619 OLIG2:3720 NTS:4057 GRP:5654
C1.spinal.cord	0.02965415	171	1.847e−01	4.844e−01	ELOVL7:102 PAQR6:128 SLC44A1:340 TBC1D12:468 SLCO1A2:491 CLDN11:637
cerebellar.hemisphere	−0.01831494	459	1.893e−01	4.844e−01	ANKRD34C:21 SNAP25:47 MINAR1:74 SSTR4:86 KIF26B:103 PCLO:135
lung	0.03405899	139	1.686e−01	4.844e−01	WNT2:126 GGTLC1:135 MSR1:303 ICAM1:305 SUSD2:329 DNASE2B:433
nucleus.accumbens	−0.03809394	93	2.063e−01	4.844e−01	DACH2:4 HTR1D:32 FAT3:121 NECAB2:148 NWD1:255 NKAIN4:404
pituitary.gland	−0.03014053	202	1.438e−01	4.844e−01	DNAH5:34 VWA7:286 RIMBP2:464 C2orf50:473 MYO15A:502 JAKMIP3:638
skeletal.muscle	0.02620559	232	1.742e−01	4.844e−01	ANO5:1 CA3:229 CKM:247 EYA4:259 FABP3:324 PRKCQ:332
substantia.nigra	0.11032107	11	2.054e−01	4.844e−01	C10orf105:201 RET:699 DBX2:1967 SLC6A11:5137 KLHL1:5421 SLC6A3:5593
uterus	−0.05527197	53	1.650e−01	4.844e−01	MRGPRF:26 TCF23:275 LCN6:643 DCHS1:861 GPR161:1742 TMEM200A:1815
eye.development	0.05178611	57	1.775e−01	4.844e−01	PROX1:1087 KLF5:1164 WNT9A:1455 FOXC1:1683 WNT5A:1850 TYR:1918
breast	0.09053765	14	2.411e−01	5.231e−01	LMX1B:45 FABP4:919 GSC:1008 LEP:1074 HCAR1:4307 ADIPOQ:4969
ectocervix	−0.06765095	25	2.422e−01	5.231e−01	MATN2:774 ADRA2A:862 COL27A1:1344 RIPOR3:1552 F10:1673 KCNK6:2321
prostate	−0.05070302	37	2.867e−01	5.734e−01	ANO7:230 LMAN1L:266 HOXB13:317 CREB3L4:767 CDH26:1060 ARHGEF38:1531
suprapubic.skin	0.02361336	181	2.776e−01	5.734e−01	TLCD3A:65 CXADR:154 COL17A1:203 CASP14:360 IL1RL2:402 SERPINB12:547
tibial.artery	−0.02833719	101	3.275e−01	6.316e−01	SUSD5:88 CSPG4:96 ACTA2:221 NKD1:346 FOXC2:505 PDE3A:591
greater.omentum	0.04602941	32	3.682e−01	6.857e−01	GFPT2:525 HBEGF:710 FABP4:919 LEP:1074 IL6:1414 MTARC1:1631
transformed.skin.fibroblast	0.01477834	243	4.332e−01	7.547e−01	ADAM9:47 COL1A1:84 CALU:133 IL1R1:304 CYP19A1:362 FBN1:408
transverse.colon	−0.01940769	144	4.248e−01	7.547e−01	SLC39A5:80 MOGAT3:90 SLC17A4:184 TRIM15:203 B3GNT6:228 LRRC31:734
Brodmann.area.24	−0.03928541	29	4.646e−01	7.603e−01	NCS1:119 PTER:571 CSPG5:703 SLC39A12:1787 NPTXR:2027 CNTN5:2787
stomach	−0.02353255	84	4.578e−01	7.603e−01	CAPN8:12 MYH13:23 VWA2:310 RFX6:587 ESRRB:775 CAPN9:1030
atrium.auricle	−0.02082643	89	4.990e−01	7.925e−01	ACTC1:64 SCN5A:92 MYBPHL:113 FSD2:208 LMAN1L:266 ADAMTS7:366
aorta	−0.01388538	151	5.589e−01	8.045e−01	SUSD5:88 ADAMTSL3:126 C1QTNF8:323 PCDH11X:442 LTBP1:601 OGN:654
cortex.kidney	0.01639123	110	5.547e−01	8.045e−01	SLC12A1:134 SLCO4C1:204 PDZK1:383 HOXB6:690 CLDN16:800 FABP1:935
esophagus.muscularis.mucosa	0.03387283	26	5.504e−01	8.045e−01	JPH2:321 NKX6−1:1343 TPM1:2109 RGS2:2241 KCNMB1:3564 GADL1:4083
peyers.patch	0.01317119	162	5.661e−01	8.045e−01	SULT1A2:8 SI:177 TMEM37:278 TMIGD1:293 ACE:403 SLC13A2:503
leg.skin	0.01062522	209	6.004e−01	8.313e−01	SPTLC3:186 COL17A1:203 BLMH:308 CASP14:360 IL1RL2:402 SERPINB12:547
hypothalamus	−0.01723883	58	6.507e−01	8.784e−01	PCDH15:43 AIRE:154 HAP1:401 PCDH19:405 KISS1R:483 DLX2:989